

Farm-to-Fortune (Agri-Edge)

An on-device AI and IoT hardware ecosystem designed to automate crop grading, prevent post-harvest cold-chain losses, and ensure fair APMC mandi pricing for farmers.

Target: Hackathon Evaluators & Judges | Prototype Status: Frontend Live Demo Ready | Date: August 2026

1. PROBLEM & SOLUTION OVERVIEW

The Core Challenge in Agriculture:

- **Manual & Biased Crop Grading:** Mandi traders and middlemen rely on subjective visual estimations. Even Grade-A quality produce is frequently categorized under Grade-C to suppress payouts to farmers.
- **Lack of Certified Quality Proof:** Smallholder farmers lack digital, tamper-proof evidence of produce quality at the farm gate.
- **Transit Spoilage (25–30% Losses):** Unregulated temperature and humidity levels during transit lead to massive post-harvest wastage.

The Agri-Edge Solution:

- **Edge-AI Grading Booth:** On-farm optical booth with camera + local TinyML model for internet-independent, real-time (< 200 ms) crop defect classification and grade verdict (Grade A/B/C).
- **Sensor Fusion:** Capacitive moisture sensors and DHT environmental sensors evaluate produce hydration & transit risk.
- **Tamper-Proof Digital QR Certificate:** Generates a verifiable QR code containing Grade verdict, moisture %, timestamp, and cryptographic verification hash.
- **Cold-Chain Telemetry & Mandi Match:** Continuous IoT tracking of vehicle transit conditions linked with real-time APMC Mandi rates to deliver +28% higher farmer margins.

2. WHAT WE HAVE BUILT (CURRENT INTERACTIVE FRONTEND PROTOTYPE)

To provide judges with an intuitive, interactive live demo during the hackathon, we developed a production-ready Web Application in `index.html`.

Technology	Implementation Role	Purpose & Impact
HTML5 & Vanilla JS Engine	Zero-dependency state engine	Handles views, multilingual switching, crop selection, modal controls, and audio synth.
Tailwind CSS (CDN)	Custom dark-emerald styling	Delivers modern dark UI (#080d0b), card overlays (backdrop-blur-md), and scan keyframes.
Lucide Icons Library	Vector icon rendering	Displays icons for hardware nodes, AI cameras, mandi locations, and QR codes.
Chart.js (CDN)	Telemetry line graph	Plots 12-hour real-time Temp (°C) and Humidity (%) curves with emerald gradients.
Web Audio API	Browser sound synthesizer	Generates laser scan sweep sounds and button feedback without external audio files.

3. WHAT REMAINS TO BE BUILT (TECHNICAL ROADMAP & BACKEND PIPELINE)

The roadmap below outlines the hardware, AI model training, and backend cloud infrastructure required to transition from the hackathon prototype to a deployed hardware system.

Domain	Pending Development Work	Target Phase
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Hardware & Embedded	Fabricate acrylic optical booth with uniform LED ring lights; program ESP32-CAM/S3 firmware for sensor reading (ADC/I2C) and camera frame buffering; assemble 18650 Li-Ion solar power circuit.	Post-Hackathon Phase 1
Edge AI & Vision	Collect and annotate 5,000+ crop defect images; train MobileNetV3/YOLOv8-nano models; quantize weights from FP32 to INT8 using TFLite; deploy onto ESP32-S3 hardware for <200ms local inference.	Post-Hackathon Phase 2
Cloud & Connectivity	Deploy Mosquitto MQTT broker on AWS/GCP for GSM telemetry streaming; build Flask/Node.js REST API server; set up PostgreSQL database tables; integrate live Government Agmarknet APMC APIs.	Post-Hackathon Phase 3

4. TEAM ROLE BREAKDOWN MATRIX

Team Role	Delivered in Prototype	Future Production Scope
Hardware Engineer	Sensor UI gauges, vehicle tracker stats, hardware latency telemetry badges.	Physical booth build, PCB design, sensor ADC calibration, battery management.
Edge AI Engineer	Bounding box overlay, defect segmentation UI, multi-crop defect metric calculations.	Model training on 5,000+ crop images, INT8 quantization, TFLite Micro deployment on ESP32-S3.
Software Engineer	Fully functional Web App (index.html) with interactive features, Chart.js, and presentation mode.	Flask/Node backend, PostgreSQL database, MQTT telemetry broker, live Agmarknet API integration.